

Auto Serendipitous Raga Generator and Recommender using AI techniques

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Abstract— Raga Alchemy is an AI Powered Platform which helps in learning, creativity and exploration of Indian classical ragas by merging AI with musicology. An Auto Serendipitous Raga Generator and Recommender automatically creates authenticated raga-based compositions. A variety of learning resources are created for novice as well as advanced learners. Recommendations of raga with respect to mood can identify emotional state of a person using raga music for AI. Not only does it help in learning and therapy, but it also allows for digital preservation of Indian Classical music ensuring cultural sustainability. The United Nations Sustainable Development goals; target 3 which promotes good health and well being. The United Nations Sustainable Development goals; target 9-encourages innovation through using artificial intelligence and deep learning for music generation and recommendation. The United Nations Sustainable Development goals; target 11-safeguarding intangible cultural heritage. It improves accessibility and inclusivity in classical music learning. The overarching study suggests that it is possible to marry tradition and technology to make Indian classical raga music globally accessible, engaging and sustainably.

Keywords— *Indian Classical Ragas, Auto Serendipitous Raga Generator, Raga Recommendation System, Musicology and AI Integration, Digital Preservation, Intangible Cultural Heritage, Music Therapy, Accessibility in Music Education.*

I. INTRODUCTION

Indian Classical Music (ICM) uses numerous elements of music in its tradition, for instance, melodic ragas to represent expressiveness, structure, and artistic freedom. The system contains several repeating rhythmic cycles from the taals and improvisational patterns in the performance. Raga Alchemy uses advanced artificial intelligence techniques which create a smooth integration of established music theory and deep learning; however, traditional mentorship contains slightly more rigid guru-disciple frameworks. Numerous melodic and time-based conventions give the raga shape outlining areas where emotional mood turns to sound and strict rules change to creative expression. Many digital music training platforms have incorporated many types of artwork during the last several years, but not many of their digital tools represented traditional melodicism or emotions accurately. There is a range of data points providing the output shape of where music theory becomes sound with input and creating shape of melody made out of sound.

Several studies, involving deep learning models such as recurrent neural networks and transformers [1], are among the most effective research globally. Researchers explored emotion aware music generation due to their fascination with the way AI systems capture and reflect emotional intent. The research involves Hindustani music, and the human AI interaction studies reflect the artist's idea of the connection between AI assisted composition and raga authenticity [4]. The structured and challenging Indian classical music contains various elements of research,

such as raga generation and recognition, as well as the principles, for instance, classification and automated learning [7]. While these works showed numerous elements of support in analysis and learning, for instance, individual tasks to represent a complete learning experience. The research contains several repeating themes from the digitalization of Indian classical music education and the traditional guru shishya system. Researchers used modern technology which created improved accessibility and a smooth transition; however, the traditional system contains slightly rougher underpaintings. Together, these studies give the platform shape outlining areas where learning turns to creativity and analysis changes to emotional understanding together.

Raga Alchemy by building on earlier ideas implies the elements and principles of art such as creativity and analysis. The system has a raga-based MIDI composition module [3] to indicate engagement and accessibility and a reasonable amount of emotion modeling for mood-based suggestions. It comprises real-time raga and swara detection with helpful feedback occupying the most prominent space and core raga principles like arohana and avarohana, both being at the top. Through the pakad/samay theory grounding of the platform, the system applies the principles of art, for instance, weight distribution. The raga exerts weight on a single side of the melodic structure; however, there is still balance despite the theory's positioning being grounded in tradition. The study emphasizes the collaboration between humans and artificial intelligence, and the emotional depth falls directly in the users' center of vision. The pakad and samay make a subtle expression challenging to capture in most digital tools and platforms. The less visually exciting areas are the underlying algorithms since they seem faded, lacking distinct features or information.

II. LITERATURE REVIEW

The process of generating music using AI and deep learning models is one of the most unique developments in the world today. Researchers have developed such a system because they are curious about how organized tunes have entered digital systems. Generative modeling does involve deep networks [2] and can be conceived over different generative approaches including variational inference, open-endedness and the creative AI [3]. The use of the direct raga particular rules defines the researchers' views on the relationship between digital grammar and Indian Classical Music. A fine project of music generation requires various categories of art like arohana, avarohana and characteristic phrases along with the principles e.g. identity of raga and consistency of classical. Various research elements in artificial intelligence are applied in certain survey studies. For example, strengths and limitations are discussed to depict

progress and existing techniques [2]. The picture of the field contains numerous repeated observations concerning the cultural roots as well as the absence of systems in the background. Judging from our project, the raga and emotion mapping approach was smooth and slightly intuitive expressions but earlier systems in Western Classical, piano based music [5], [6] raga and emotion were found with slightly different musicality. There are several associations that influence the emotional intent which determines the locations wherein a user's feeling becomes ragas and other moods commonly associated with it. The project utilized realistic amounts of technical support for interactions. Furthermore, Hindustani music's human-AI interaction study incorporates creative aspects such as collaboration and experimentation [4].

In contrast, system usability features functional aspects, for example, learning and companion support. Also our work provided structure for the learning process and also for the usability process. Our project has made by the idea of using ai in the music to make our indian classical music more fun and accesible to all the audience and not just as tutor and also taking realtime feedback to improve learning experience.

Previous studies related to creating music based on the raga format, computer-based musical education, and automating teaching have mostly concentrated on particular aspects of Indian classical music, such as how to classify different types of music and how to identify categories of raga's music[7][9]. This study aims to develop a complete picture where all parts of creating your own raga, getting emotional recommendations based on what you're listening to (i.e., what kind of mood you're in), being able to detect swaras (notes), create interactive lessons with videos, and use them. In doing so, this project will allow those new to Indian Classical Music (ICM) to obtain an understanding of ICM through a more complete and efficient means, while also integrating with each other to be practical, simple and successful in both learning and enjoying making music in ICM[10][11].

III. METHODOLOGY

Raga Alchemy's methodology is an extremely comprehensive framework and is one of the most advanced worldwide. The creation of Raga Alchemy results from a desire for a deeper understanding of the combination of data-driven machine-learning techniques with ICM's domain expertise of the developers. There are multiple stages within the full flow of the system, with the developers' basic concept of Musical Authenticity in relation to User Interaction forming the basis of how each step connects to the previous and subsequent steps. The Raga Alchemy project includes an excellent integrated system that provides users with an emotional raga recommendation and the ability to create music based upon this emotion, while also utilizing the principles of real-time raga detection/swara detection/usage, etc.

A. User Input and Data Pre Processing

The developers obtained their various sources of audio data within Raga Alchemy based upon popular and well-known databases of audio for Indian classical music, to provide authenticity, variety and structure. In addition to this, much of the audio contained in the full collections of recordings used in Raga Alchemy was previously found in CompMusic (Dunya) datasets, that have been extensively

used previously for studies of raga recognition and classification through comparisons with one or more raga classifications [7]. The choice of curated recordings produced by the developers allowed for a smooth training experience, as well as a slight glossing/flawless final audio. For one thing, there are a few less-tidy labelled facial emotion databases in contrast to those used for emotion detection. This is to say, there are many recordings used to shape the system, wherein areas indicate that emotion-based music generation follows earlier works [5].

Secondly, the treatment of the Raga Alchemy system has a tonic normalisation of pitch variance as compared to previous examples in the literature of raga recognition and classification studies [11]. Also, the developers made use of realistic amounts of audio samples for the purpose of extracting features. This involved several layers of obfuscated data from the Librosa library - those created will appear technically completed because of the double denim texture. The methodology for Raga Alchemy follows a structured shape in that the developers have applied technical principles to form the system. In conclusion, both data preparation and processing methods of Raga Alchemy ensure that it remains technically sound while still conforming to the established principles of AI Based music analysis and generation also.

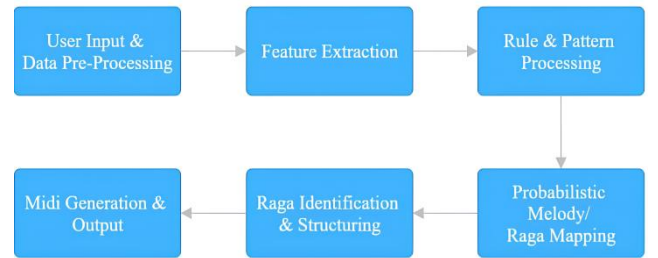


Figure 1: Workflow of Raga Alchemy

We created this system out of a sheer desire to create a significant connection between the way in which a user provides an input to create a musical output. This program uses extensive data collection and analysis in order to produce real ragas, which reflect the creator's vision of combining classical music theoretical concepts with digital technology. Each raga created, which is exquisite in both nature and sound, contains many elements of the art form, including melody, rhythm, and tone, as well as principles such as authenticity, both individually and collaboratively.

The system used numerous stages of analysis in Raga Alchemy, for instance, rules and patterns to represent structure, context, and musicality. The workflow contains several repeating steps from the feature extraction and the raga mapping in the background. Raga Alchemy uses machine learning which created a smooth melodic path and slightly nuanced associations; however, the early stage contains slightly simpler data preprocessing. Numerous musical constraints give the output shape outlining areas where pitch turns to rhythm and raga identification changes to a composition. Moreover, the structure of the Raga Alchemy contains machine learning methods such as probabilistic mapping, while the background contains statistical methods, for instance, Indian classical music associations.

The scale and proportion of the Raga Alchemy can be explained using MIDI generation, which possesses a similar manipulation. The latter results in the generation

of the composition since it has a massive amount of data to structure presenting an authentic figure. The Raga Alchemy workflow by the system implies the elements and principles of music such as raga identification and MIDI generation. The output has digital files to indicate authenticity and validity and a reasonable amount of probabilistic AI techniques

B. Feature Extraction

Feature extraction, occurring once the input has been cleaned and prepared, is among the most significant stages globally. The system computes features for audio recordings due to its fascination with the way sound appears in spectral descriptors. The process involves MFCC and chroma representations, and the melodic progression of the signal reflects the system's idea of the connection between tonal structure and harmonic distribution. The accurate and precise feature extraction contains various elements of sound, such as pitch contours, as well as the methodologies, for instance, in music information retrieval research [9], [10].

For the music generation component, numerous elements of symbolic representations are used, for instance, MIDI encoded note sequences to represent swaras, note durations, and tempo. The system contains several repeating patterns from the note sequences and the structured format in the background. Deep learning based music generation research [1], [3] used RNN and VAE models which created a compact yet descriptive feature set and slightly structured processing; however, the subsequent identification contains slightly more reliable rule-based processing. Numerous machine learning models give the system shape outlining areas where audio signals turn to features and probabilistic raga mapping changes to MIDI generation.

C. Rule and Pattern Processing

Moreover, the framework of the Indian Classical Music contains strict theoretical rules such as arohana, avarohana, and pakad, while the module contains time based associations, for instance, Hindustani music. The system used realistic quantities of predefined raga rules for maintaining authenticity. The earlier studies on human AI interaction [4] comprise layers of data driven systems that are slightly inconsistent, contributing to the musically inconsistent outputs of the generated music. Rule enforcement has a critical role measuring against theoretical frameworks as the system applies musical principles to deliver an authentic piece.

To address this limitation, classical music theory is applied to the processing pipeline, for instance, rule aware processing. Despite the fact that detecting and generating are very much weighted to one side of the structural grammar, the respect the Raga Alchemy system has for raga identity has not been lost due to the cultural context of the raga output directly aligned with the individuals in the audience who have a raga identity. The cultural foundation of the Raga Alchemy system has also been emphasized through its avoidance of generic, AI-generated melody production [1], [2] while still being consistent with the musical output of individuals who have a raga identity. The less musically consistent areas are the data driven systems since they seem faded, lacking distinct features or information.

The system identifies characteristic phrases and recurring melodic motifs due to its fascination with the way ragas define specific melodies. The module involves a bridge between traditional musicology and modern artificial intelligence, and the melodic tendencies of the system reflect the developer's idea of the connection between music and cultural integrity. The excellent and consistent workflow contains various elements of generation, such as motifs and patterns, as well as the principles, for instance, authenticity and cultural integrity.

D. Probabilistic Melody and Raga Mapping

The system uses numerous elements of deep learning in raga classification, for instance, CNN and LSTM architectures to represent local spectral patterns and long-term melodic relationships. The architecture contains several repeating features from the spectral patterns and the melodic relationships over time.

This approach is inspired by earlier classification techniques which created a effective recognition [10] and [11]; however, the generation module contains sequence modeling techniques similar to those used in RNN and VAE based systems [1], [3]. Numerous constraints give the sequences shape outlining areas where probabilistic learning turns to theoretical enforcement and creative paths change to authenticity.

E. Raga Identification and Structuring

Moreover, the identification stage of the system contains probabilistic analysis, while the structuring stage contains rule-based validation. The model used realistic quantities of learned representations for raga detection. The output comprises layers of processed features that are slightly thick, contributing to the transparent appearance of the system's performance certainty [10], [11]. The identification has a structured space measuring confidence scores as the system applied evaluative principles to deliver an excellent piece. The output also contains a sense of reliability since it is angled on predicted labels and confidence values. The process seems to consist of three stages.

Also about the identification and structuring the use in our project of the indian classical music we have use various movements various variations for getting better fit for our project. this generation mode uses raga grammar which creates a consistent transition and slightly authentic sequences; however, earlier raga generation studies [7] often demonstrated sequence creation but lacked deeper structural enforcement. Numerous validations give the melody shape outlining areas where note transitions turn to traditional conventions.

F. MIDI Generation and Output

Moreover, the representation of the MIDI conversion contains playable pitch values such as durations, instrument parameters, and notes, while the symbolic representation contains standard formats, for instance, [1] and [3]. The system used realistic quantities of structured encoding for flexible musical manipulation. The melodic structure comprises layers of pitch values that are slightly thick, contributing to the glowing appearance of the system's output. This MIDI representation has a symbolic shape measuring note durations as the system applied

deep learning principles to deliver an excellent piece. The module applied the principles of user engagement while producing MIDI files, for instance, visual presentation. The system exerts weight on a single side of the playback; however, there is still balance despite the visual presentation being on-platform. The module emphasizes the raga's theoretical rules since it is visual, and the swara sequence falls directly in the learners' center of vision. The playback and the visuals make a subtle learning resource challenging to capture in most automated tutoring systems [9].

The scale and proportion of the MIDI generation can be explained using different parameters, which possesses a similar manipulation. The latter results in a creative artifact since it has a massive tempo to sequence length presenting a learning resource. This module implies the elements and principles of music such as analysis and feedback. The MIDI generation has structured encoding to indicate creativity and a reasonable amount of playback for music, exerting weight on one side, thus making it important for playback music to help the user to sing without the help of others instrumental players.

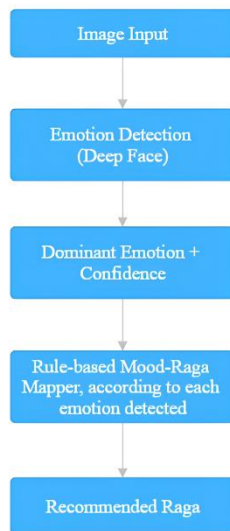


Figure 2: Workflow of Emotion to Raga Mapping

Fig. 2 shows the workflow for our Raga Alchemy project, which contains three main features. This diagram outlines the first feature of our project, emotion recognition. This feature is one of the main components that support the two other features. To use this feature, the user will upload an image and then through deep face analysis relational models will analyze several aspects of the user's face and determine the emotions the user is displaying in the image. Once the emotions associated with the user's image are identified, they are connected to probabilities based on the image analysis. The image analysis is used to make predictions of specific ragas to be played based on the emotions found in the user's image. The user can submit many images of themselves or of someone else from whom they would like to have their emotions predicted and would like to know which raga to be played based on their emotions.

Once dominant emotion is received it is sent to Mood Raga Mapper through rule-based method designed to map each validated emotional category to a specific raga following predetermined rules based upon musicology set out in this module. After calculating a recommendation based on rules applied, system outputs a recommendation for which the user can be provided with this data as well as raga value associated with it comprised from user

settings providing minimum of one measure per raga presented based upon its emotion and use with other emotions. These ragas are expected to conform aesthetically and formally to both classical aesthetic requirements and musical construction standards.

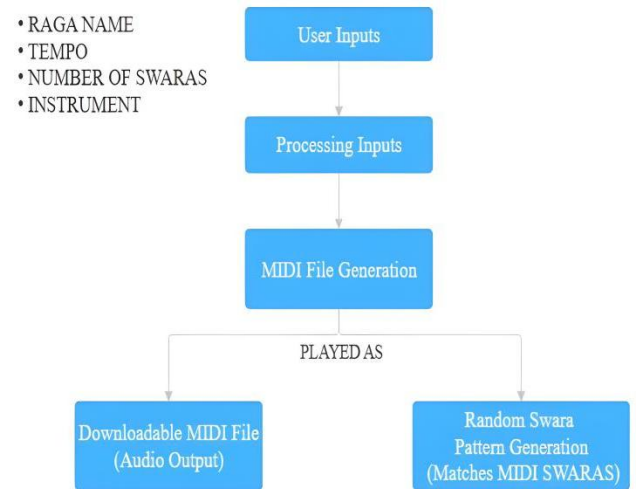


Figure 3: Workflow of MIDI Generation

Fig. 3 provides a visual representation of the workflow for creating a unique MIDI file using the MIDI generation tool. A MIDI file is a music file containing both musical instruction (performance) and audio specific to the instrument being played (i.e., piano, guitar, trumpet) during singing or practicing. When you create a MIDI file using the MIDI generation tool, you initially have to select a raga to base your file on, as well as which instrument you would like to create your file with. After making selections, you will then choose the tempo for your MIDI file (i.e., fast or slow). The tool will then create a MIDI file based on your selections, providing you with swaras as well as a downloadable version of your MIDI file. The MIDI generation tool is a great resource for many users, allowing them to access the same MIDI files when practicing.

The MIDI file is generated after the pre-processing phase. Therefore, the rules of the raga will determine the sequence of sound generated by the MIDI notes, which are a mapping of the swara frequencies. The MIDI notes correspond to the mapped sequence of swaras played together to create the relative pitch of that scale. The time values of each note will also be ascribed as per the tempo of the music. A technique is used to align the durations of notes by tempo scheduling. The output will be a standard MIDI file with pitch, timing, and instrument information included. In essence, this is the representation of two equal parts.

There are two ways the generated MIDI file can be utilized. The user may either download the audio directly or listen to the rendition of their composition created by the MIDI. Additionally, random swara pattern generation may be achieved through the system's random pattern generator.

All of the swara patterns generated will utilize the same swaras of the raga to be rendered in their pattern composition. Consequently, all minimum phrases required for fitting within raga compositions will result in modification of the preceding phrase to comply with the rules

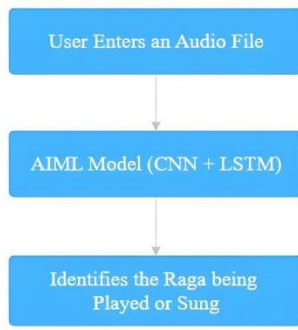


Figure 4: Workflow of Advanced Raga Detection

The workflow in Fig. 4 illustrates how to implement advanced methods for raga detection in the Raga Alchemy tool. The feature uses an mp3 vocal file of one or more specific singers to help determine if the music in the mp3 file has been performed accurately with reference to the raga used by the singer. The workflow provides outputs displaying the detected ragas and their associated confidence scores based on the analysis performed. The function of this feature is to input an audio file and then apply CNN and LSTM networks to perform the analysis.

The process defined in Fig. 4 has successfully documented recognition whilst retaining significance regarding Classical Indian Music necessary for generations to inform those people to come. Signal processing paired with deep learning has allowed for computational analysis of Classical Indian music using both LSTM's(CNN) and convolutional networks to identify which Ragas belong to an audio file. Thus providing users with a confidence score to understand how accurately/closely an audio file represents the original Raga that should be sung. This capability provides a method for practitioners of Indian Classical Music to evaluate their own skills versus the Raga(s) they are working on. Our goal is to demonstrate the value of classical music through these types of tools while allowing users to engage with the variety of educational materials available on our website compared with traditional methods of learning via gurus.

Sr. no.	Western Notation	Indian Notation	Frequency Ratio(natural)
1	C	Sa	1/1
2	C#	Re	16/15
3	D	Re	9/8
4	Eb	Ga	6/5
5	E	Ga	5/4
6	F	Ma	4/3
7	F#	Ma'	7/5
8	G	P	3/2
9	Ab	Dh	8/5
10	A	Dh	5/3
11	Bb	Ni	9/5
12	B	Ni	15/8
13	C'	Sa' (Next octave)	2/1

Table 1: Western-Indian Musical Note Mapping

In Table 1, you will notice the different mappings of the western music system to the Indian music system, as well as the various frequency ratios for each of the individual mappings and notations. You can see that in this example, that any of the musical notes fall into their respective mappings with their corresponding frequency ratios (i.e., Sa = 1/1 = C). Similarly, the other musical notes have specific western and Indian associations along with the respective frequency ratios (i.e., Re = 16/15 = C#, 9/8 = D). All other notes are associated or identified in a similar

manner with their respective natural frequency ratios. For example, you will see that the Ma - Shuddh frequency ratio is 4/3 and the Ma - Tivra frequency ratio is 7/5. The separation between the two different musical notes is evidenced by the difference in the frequency ratio. This table can be useful in the analysis of swaras, creation of midi files and also in the identification of ragas.

IV. RESULTS

A. Implementation Stack & Tools

1. Frontend (User Interface) -

React.js → API Calls → Backend Services

React.js: Modern web interface

CSS with Indian Theme: Saffron, green, maroon color scheme

Real-time Updates: Live feedback and progress indicators

File Upload Handling: Image and audio processing

2. Backend (AI Brain) -

Flask API → AI Models → Music Processing → Response

Python Flask: Web server and API endpoints

TensorFlow/Keras: Machine learning framework

Librosa: Audio analysis and processing

PrettyMIDI: Music generation and MIDI creation

DeepFace: Emotion recognition from images

3. AI/ML Core -

Input Data → Preprocessing → Model Inference → Results

Pre-trained Models: Emotion detection, audio feature extraction

Custom Algorithms: Raga composition rules, swara detection

Real-time Processing: Immediate analysis and generation

B. Module Results

Here are the results of feature no. 1 which is emotion to raga mapper :

No	Images	Emotion Detection	Confidence Score	Raga Recommended
1	Image 1	Surprise	84%	KEDAR, SHREE, MALKAUN S
2	Image 2	Neutral	84%	YAMAN, KAFI
3	Image 3	Peaceful	92%	BHAIRAVI, AHIR_BHAI RAV
4	Image 4	Romantic	85%	KHAMAJ, KAFI
5	Image 5	Neutral	88%	BAGESHRE E, YAMAN

Table 2: Results of Emotion To Raga Mapper

The Table 2 above shows the test on the first feature, which is to map emotions to ragas. This feature allows a user to upload an image of their emotion, and then the user will be suggested a specific raga based on their

emotional state. The user will then be able to listen to that classical raga according to their mood or emotion. As can be seen from this table above, we tested this feature by providing five different images as input to determine which emotion(s) are being detected and at what level of confidence as well as the ragas that have been suggested based on those emotions.

Images 1 through 5 demonstrate how different emotions were identified from specific input images, including: 1) Emotion: Surprise (84% confidence), recommended ragas: Kedar, Shree, Malkauns; 2) Emotion: Neutral (84% confidence), recommended ragas: Yaman, Kafi; 3) Emotion: Calm (92% confidence), recommended ragas: Bhairavi, Ahir_Bhairav; 4) Emotion: Romantic (85% confidence), recommended ragas: Khamaj, Kafi; and 5) Emotion: Neutral (88% confidence), recommended ragas: Bageshree, Yaman. Each image reflects a distinct emotion and provides additional context through the recommendation of associated ragas.

The Mood-Raga Mapper created understandable pairings of emotion profiles (the identified feelings that correspond to ragas) with ragas using the foundational aspects of Indian classical music such as the arrangement of swaras, the idea of rasa, and long-standing cultural conventions regarding how to play ragas. As an example, Kedar, Shree, and Malkauns are related to the emotion of Surprise because they each have extremely rapid changing tonalities in their respective compositions and all demonstrate unique aspects of how a melody progresses, which causes Surprise which is an expression of unpredictability and depth. Neutral was mapped to Yaman, Kafi, and Bageshree because they have an equal number of notes distributed evenly through the respective swaras and are often used in situations to provide general listeners with a calming experience; these ragas provide emotional stability. The mappings of Peaceful to Bhairavi and Ahir Bhairav show that the use of komal swaras provide a smooth transition thus allowing for calmness or devotion to be expressed. Finally, Romantic was mapped to Khamaj and Kafi ragas; both are associated with the shringara rasa (romantic) emotion by having a subtle but very expressive melodic composition. By systematically comparing all of these pairings (emotional profiles to ragas) with a strong musicological basis, the mapper has established both accurate and transparent interpretability and created an educational bridge between individuals and how they interpret their feelings.

The results of feature no. 3 Advanced Raga Detection :

No.	Audio Files	Raga Detected	Confidence Score
1	Audio 1	ASAVARI	91%
2	Audio 2	BHAIRAV	89.8%
3	Audio 3	TODI	97.4%
4	Audio 4	MALKAUN S	86.5%
5	Audio 5	BHAIRAVI	92.9%

Table 3: Results of Advanced Raga Detection

The results from Table 3 show the results of the third feature of the Raga Alchemy project with regards to advanced raga detection through audio file input. The audio file can contain any form of vocals or instrumentals of singing classical ragas which will be used as input, then we will check to

see if we were accurate and how accurate we were based on the confidence scores we assigned for that specific raga and the use of CNNs and LSTMs in our project. In doing this we have detected the following ragas with the following confidence scores for the first five audio files used as input: the first audio file as our input was Asavari with a confidence score of 91% therefore it is thought to be accurate 91%. The second audio that was played was Bhairav with a confidence score of 89.8%. The third audio played was Todi with a confidence score of 97.4%. The fourth audio was Malkauns with a confidence score of 86.5%. The fifth and last audio played was Bhairavi with a confidence of 92.9%. Therefore based on the above mentioned evidence we believe our advanced raga detection system is an accurate method of detecting ragas using audio files as input.

V. CONCLUSION

Raga Alchemy is a new digital platform that uses ai and the technology to teach the indian classical music in a specific and interesting way, it also helps to check if you are singing any specific classical song in a proper way or not basically acting as a tutor to keep track of your voice in a indian classical song also it has different features with which it helps the user to learn more about the indian classical traditional music. Here we blend the traditional music with the modern technology like ai and also it helps preserve the traditional music while also making it available to all the people our aim is simple to making classical music available to learn no matter who they are it's a step towards telling our classical music importance.

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